SMART INDIA HACKATHON 2026

TITLE PAGE

- Problem Statement ID – SIH26104
- Problem Statement Title- AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID-
- Team Name : AURAfox

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PROBLEM STATEMENT

1. **AI VOICE CLONING**
AI-generated voice cloning enables attackers to convincingly impersonate trusted individuals.

2. **HIGH-RISK ATTACKS**
Can lead to financial fraud, social engineering, and unauthorized actions.

3. **REAL-TIME GAP**
Existing systems lack reliable real-time detection and prevention of voice-cloning attacks.

4. **AI-POWERED NEED**
Need to detect synthetic voices, verify identity, and assess risk dynamically.

5. **SAFE DEPLOYMENT**
The solution must provide real-time alerts and preventive actions with privacy and scalability.

PROPOSED SOLUTION

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1 **LIVE DETECTION**
Built an AI-powered system that detects AI-generated and cloned voices during live calls.

VOICE INTELLIGENCE
Analyzes voice features such as sound patterns, tone, speech style, and speaker characteristics in real time.

3 **DEEPFAKE DETECTION**
Uses deepfake detection models to identify synthetic or manipulated speech.

4 **SPEAKER VERIFICATION**
Compares the caller's voice with trusted reference voice samples to detect impersonation.

5 **DYNAMIC RISK SCORE**
Generates a live 0–100 Impersonation Risk Score that updates throughout the call.

6 **THREAT FUSION**
Combines voice authenticity, speaker consistency, and other risk signals to assess the threat level.

7 **PRIVACY-FIRST INTEGRATION**
Provides privacy-focused APIs and SDKs for banking, enterprises, telecom, and government systems.

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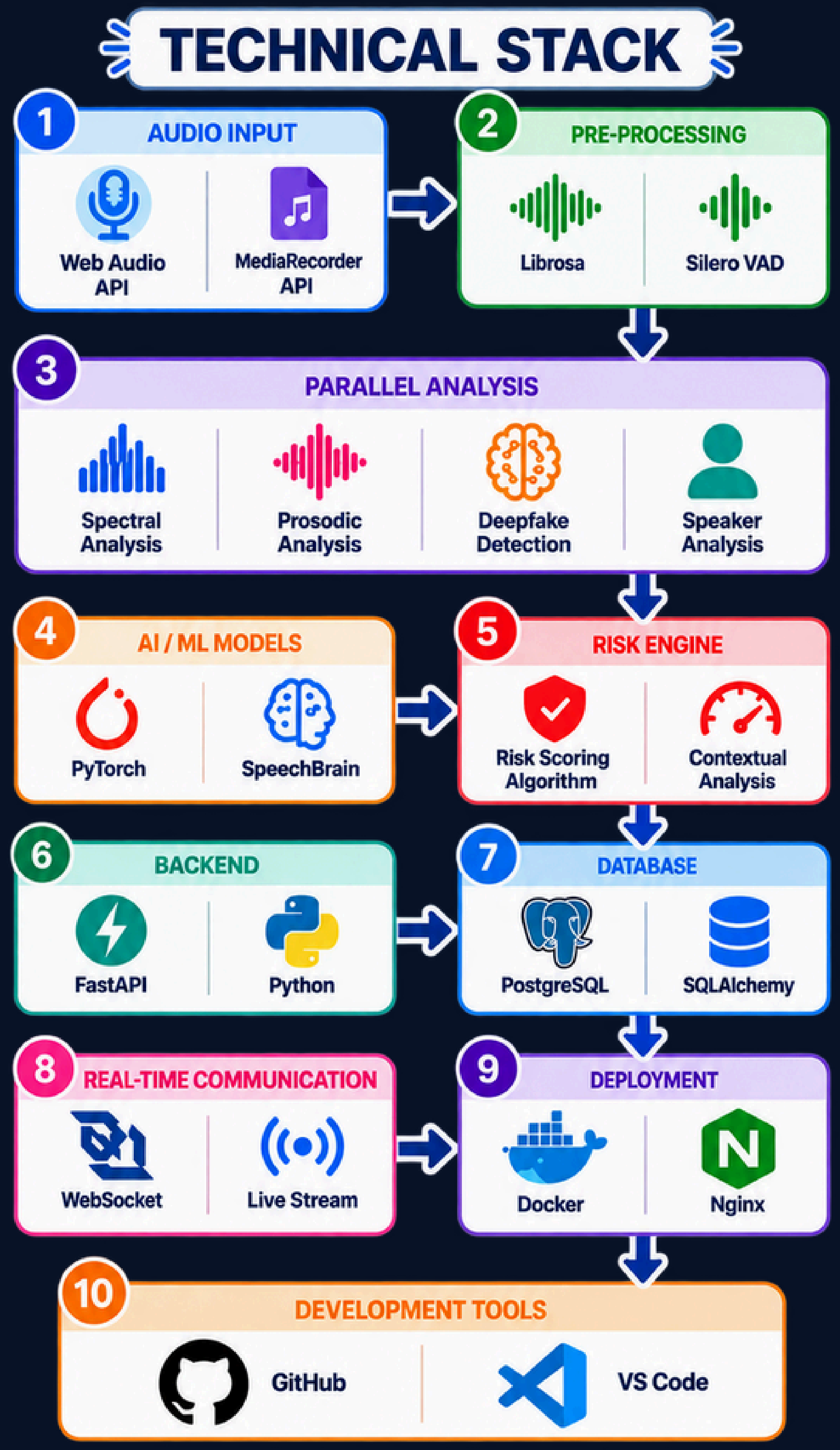
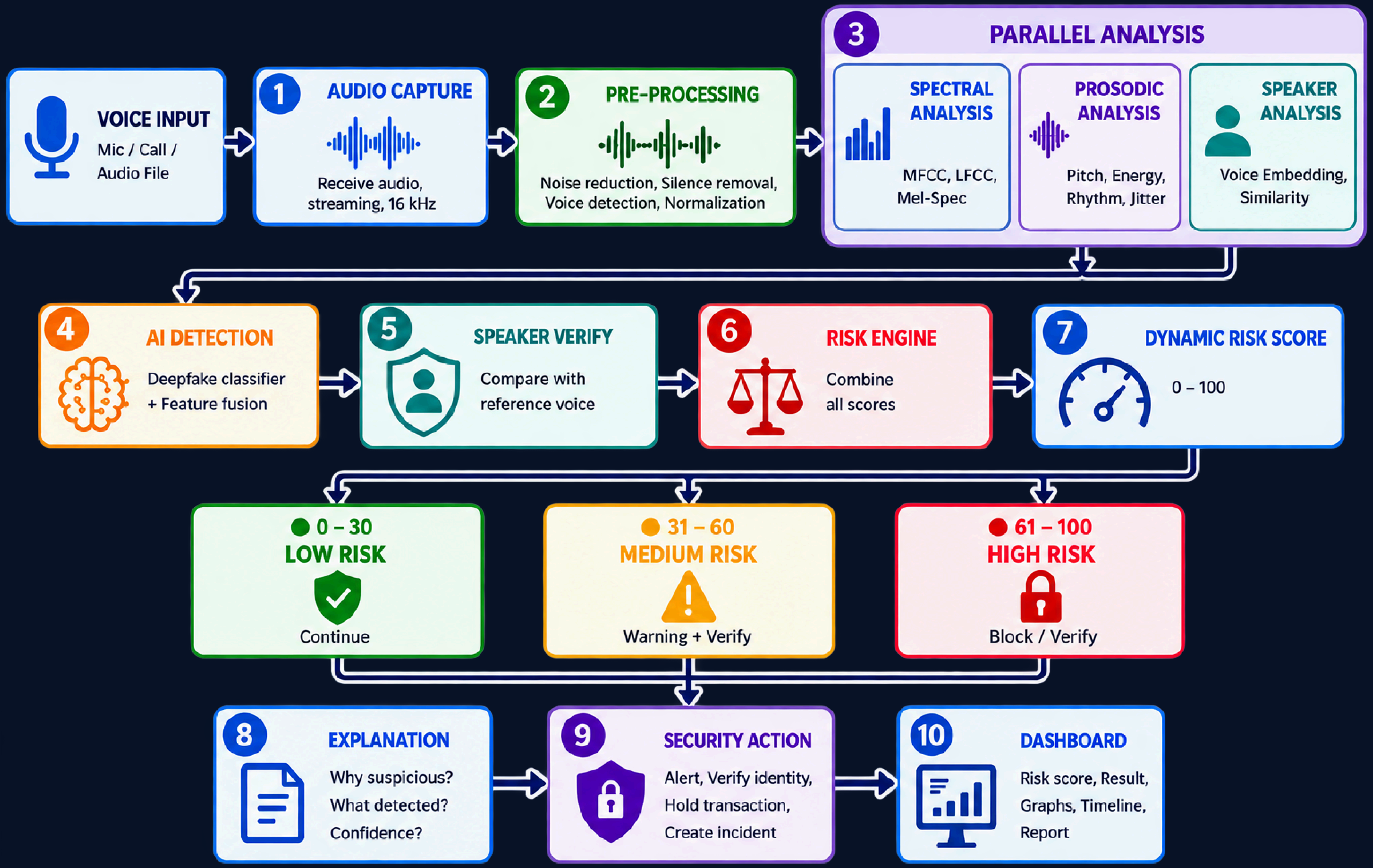
TECHNICAL APPROACH





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FEASIBILITY

EXISTING INFRASTRUCTURE

Uses existing microphones, VoIP and communication infrastructure; no specialized hardware is mandatory.

01

REAL-TIME PROCESSING

Audio can be processed in short streaming chunks, enabling near-real-time detection.

02

MODULAR ARCHITECTURE

Modular architecture allows individual AI components to be upgraded without redesigning the entire system.

03

OPEN DATASETS & BENCHMARKS

Open research datasets and established audio anti-spoofing benchmarks enable model training and evaluation.

04

VIABILITY

FRAUD RISK REDUCTION

Reduces risk of voice-cloning fraud, social engineering and unauthorized approvals.

05

CUSTOMIZABLE DEPLOYMENT

Risk thresholds can be customized for different environments such as banking, government and corporate operations.

06

INDIAN LANGUAGES & ACCENTS

Can support Indian languages, regional accents and diverse communication environments through suitable training and evaluation.

07

MULTI-SECTOR SCALABILITY

Long-term platform potential across banking, telecom, enterprises, government and customer-support systems.

08

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IMPACT & BENEFITS

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01 **F**
FRAUD PREVENTION
Detects AI-generated voices before attackers can manipulate high-risk transactions or approvals.

02 **R**
REAL-TIME PROTECTION
Provides warnings while the conversation is still happening instead of analyzing the call afterward.

03 **R**
REDUCED SOCIAL ENGINEERING
Helps identify impersonation of executives, officials, employees and trusted individuals.

04 **C**
CYBER RESILIENCE
Creates a reusable security layer against emerging AI-powered voice impersonation attacks.

05 **F**
FINANCIAL SECURITY
Provides additional protection for high-value transactions and sensitive financial instructions.

06 **P**
PRIVACY PRESERVATION
Supports edge processing and feature-level logging to minimize retention of sensitive voice recordings.

07 **I**
IMPROVED TRUST
Strengthens confidence in voice-based communication for organizations and users.

08 **S**
SCALABLE DEPLOYMENT
Can be integrated into banking, telecom, enterprise and government communication environments.

OUTCOME

- S Safer Communication
- L Reduced Losses
- O Operational Efficiency
- E Stronger Security Ecosystem

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RESEARCH AND REFERENCES

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